

Finite Primitive Sets Are Sets of Synthesis for p -Adic Unipotent Weighted Orlicz Algebras

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Abstract

Question 8.12 of Carter’s paper asks for geometric conditions on subsets of the primitive and coadjoint-orbit spectra which imply spectral synthesis for Hermitian weighted Orlicz algebras on unipotent p -adic groups. We prove that every finite subset is a set of synthesis. More generally, the result holds for every finite closed family of admissible primitive points of a locally elliptic group. Carter’s CCR theorem makes all primitive points admissible in the p -adic unipotent case. The proof combines compact-open smoothing, finite-dimensional representation corners, finite Hecke algebras, and an exact central-support cutoff in Carter’s generating ideal $j(C)$. We also retain the earlier compact-open synthesis theorem in the abelian case.

Status. Candidate substantial partial result, likely valid, subject to human review. It completely establishes the geometric criterion “ C is finite” for all groups in Question 8.12, including nonabelian ones. The source question is open-ended, and infinite subsets of nonabelian orbit spaces remain untreated.

1 Source question

The source is Max Carter, *Weighted Orlicz $*$ -algebras on locally elliptic groups*, arXiv:2503.20735v2, Question 8.12 on page 29 [1]. Given a p -adic unipotent matrix group N , Carter identifies

$$\mathrm{Prim}_*(L^\Phi(N, \omega)) \cong \mathfrak{n}^* / \mathrm{Ad}^*(N)$$

and asks which geometric properties of a subset C imply $\ker(C) = j(C)$.

2 Definitions

Let G be locally elliptic, let (Φ, Ψ) be a complementary pair of Young functions with $\Phi \in \Delta_2$, and let ω be an L^Φ -weight with $1/\omega \in L^\Psi(G)$. Put

$$\mathcal{A} = L^\Phi(G, \omega)$$

and assume that $\mathcal{A}$ is Hermitian. Carter identifies the irreducible $*$ -representations of $\mathcal{A}$ with the irreducible unitary representations of G and defines, for closed $C \subseteq \mathrm{Prim}_*(\mathcal{A})$,

$$\ker(C) = \bigcap_{P \in C} P.$$

hermitian (which is the case if ω is sub-additive, for example) and suppose that I is a closed ideal of $L^\Phi(N, \omega)$ with $\text{hull}_*(I) := C \subseteq \text{Prim}_*(L^\Phi(N, \omega))$. Let $j(C)$ be the ideal of $L^\Phi(N, \omega)$ as defined in Theorem 7.1. Then, the ideal I must sit between $\ker(C)$ and $j(C)$ i.e. $j(C) \subseteq I \subseteq \ker(C)$. In particular, understanding the closed ideals of $L^\Phi(G, \omega)$ with hull equal to C would require one to understand the ideal theory of the algebra $\ker(C)/j(C)$.

The set $C \subseteq \text{Prim}_*(L^\Phi(N, \omega))$ is called a **set of synthesis** if $\ker(C)/j(C)$ is trivial, or equivalently, $\ker(C)$ is the only closed ideal of $L^\Phi(N, \omega)$ with hull C . The property of $L^\Phi(N, \omega)$ being Wiener is equivalent to the empty set being a set of synthesis, so under our assumption that $L^\Phi(N, \omega)$ is Hermitian, it follows that the empty set is a set of synthesis by Theorem 1.3. Also, by the previous corollary, we may identify C with a subset of $\mathfrak{n}^*/\text{Ad}^*(N)$. We thus pose the following problem.

Question 8.12. *Let N be a group of n -dimensional unipotent matrices over a p -adic field. Let (Φ, Ψ) be a complementary pair of Young functions and ω an L^Φ -weight on N with $1/\omega \in L^\Psi(N)$ and $L^\Phi(N, \omega)$ Hermitian. Given*

$$C \subseteq \mathfrak{n}^*/\text{Ad}^*(N) \cong \text{Prim}_*(L^\Phi(N, \omega)),$$

can geometric properties of C , when viewed as a subset of $\mathfrak{n}^/\text{Ad}^*(N)$, be linked to algebraic properties of the algebra $\ker(C)/j(C)$? In particular, what geometric properties of C imply that it is a set of synthesis?*

Figure 1: Question 8.12, rendered from page 29 of arXiv:2503.20735v2.

His smooth functional calculus produces a closed two-sided ideal $j(C)$ with $j(C) \subseteq \ker(C)$ and $\text{hull}_*(j(C)) = C$; the set C is a *set of synthesis* if $j(C) = \ker(C)$ [1, Theorem 7.1].

Write $\mathcal{D}(G)$ for the compactly supported locally constant functions on G . If $H \leq G$ is compact-open, set

$$e_H = \mu(H)^{-1} 1_H.$$

Then $e_H = e_H^* = e_H * e_H$, and $\pi(e_H)$ is the orthogonal projection onto the H -fixed vectors of a unitary representation π . Call π *admissible* if these fixed-vector spaces are finite dimensional for all compact-open H .

3 Proof intuition

There are two scales in the proof. At the analytic scale, convolution on both sides by e_H approximates the identity. For a finite family of admissible representations, the smoothed algebra $e_H \mathcal{A} e_H$ is evaluated inside a finite direct sum of matrix algebras. A bounded linear lift in that finite-dimensional image corrects dense test functions so that all selected representations annihilate them exactly.

At the algebraic scale, every test function lies in a finite-dimensional Hecke algebra. If all selected representations annihilate a test function d , they also annihilate the central support projection p_d of d in that Hecke algebra. Carter's smooth functional calculus turns a scalar multiple of p_d back into exactly p_d , placing it in $m(C)$. Since $d = p_d * d$, the test function lies in $j(C)$. Passing first through the finite corner and then through the compact-open approximate identity proves synthesis. In the p -adic unipotent case, CCR makes every fixed-vector projection compact and therefore finite rank.

4 Finite Hecke support

Lemma 1 (Dense locally finite test algebra). *$\mathcal{D}(G)$ is dense in $\mathcal{A}$, and every $d \in \mathcal{D}(G)$ belongs to a finite-dimensional convolution $*$ -subalgebra of $\mathcal{D}(G)$.*

Proof. The Δ_2 hypothesis gives density of compactly supported functions in the weighted Orlicz space. Since a locally elliptic group has a compact-open basis, uniform approximation on compact sets by locally constant step functions, together with local boundedness of ω , gives density of $\mathcal{D}(G)$.

Fix $d \in \mathcal{D}(G)$. Its compact support is contained in a compact-open subgroup K . Uniform local constancy on K supplies an open subgroup of K under whose left and right translations d is invariant. Replacing it by its normal core gives an open normal subgroup $H \triangleleft K$ with the same property. The H -bi-invariant functions supported in K form a finite-dimensional convolution $*$ -algebra, isomorphic after Haar normalization to the finite group algebra $\mathbb{C}[K/H]$, and contain d . $\square$

Lemma 2 (Algebraic kernel elements lie in $j(C)$). *Let $C \subseteq \text{Prim}_*(\mathcal{A})$ be closed. Then*

$$\mathcal{D}(G) \cap \ker(C) \subseteq j(C).$$

Proof. Take $d \in \mathcal{D}(G) \cap \ker(C)$ and place it, using Lemma 1, in a finite-dimensional convolution $*$ -algebra B . Viewed in its faithful finite-dimensional regular representation, B is a finite-dimensional C^* -algebra. Let p_d be the central support projection of d in B , so the ideal generated by d is Bp_d and $p_d * d = d$.

For every irreducible representation π whose kernel lies in C , the kernel of the restricted $*$ -representation $\pi|_B$ is an ideal containing d . It therefore contains p_d , so $\pi(p_d) = 0$. Choose

$$0 < a < 2\pi, \quad a\|p_d\|_1 \leq 1.$$

Carter's cutoff Proposition 5.5 provides $\varphi \in A_\gamma$ which is zero near 0 and satisfies $\varphi(a) = 1$. Thus $\varphi\{ap_d\} \in m(C)$. Since $p_d * p_d = p_d$ and $\varphi(0) = 0$, the defining power series for the functional calculus gives

$$\varphi\{ap_d\} = \varphi(a)p_d = p_d.$$

Hence $p_d \in m(C) \subseteq j(C)$, and $d = p_d * d \in j(C)$. $\square$

5 Finite admissible sets

Theorem 3 (Finite admissible synthesis). *Let $C = \{\ker \pi_1, \dots, \ker \pi_r\}$ be a finite closed subset of $\text{Prim}_*(\mathcal{A})$. If every π_i is admissible, then C is a set of synthesis.*

Proof. Put $I = \ker(C)$. Fix a compact-open subgroup $H \leq G$ and define

$$\mathcal{A}_H = e_H * \mathcal{A} * e_H, \quad \mathcal{D}_H = e_H * \mathcal{D}(G) * e_H.$$

The density in Lemma 1 and boundedness of convolution by e_H show that $\mathcal{D}_H$ is dense in $\mathcal{A}_H$. Consider

$$T_H : \mathcal{A}_H \longrightarrow \bigoplus_{i=1}^r \mathcal{B}(\mathcal{H}_{\pi_i}^H), \quad T_H(f) = (\pi_1(f), \dots, \pi_r(f)).$$

The codomain is finite dimensional by admissibility. Therefore $T_H(\mathcal{D}_H)$ is a closed linear subspace. Density implies

$$T_H(\mathcal{D}_H) = T_H(\mathcal{A}_H).$$

Choose a linear right inverse

$$R_H : T_H(\mathcal{A}_H) \longrightarrow \mathcal{D}_H$$

for T_H . Its domain is finite dimensional, so R_H is bounded for the $\mathcal{A}$ -norm on its range.

Let $f \in I$ and put $f_H = e_H * f * e_H$. Then $f_H \in I \cap \mathcal{A}_H$. Choose $d_n \in \mathcal{D}_H$ with $d_n \rightarrow f_H$ in $\mathcal{A}$ and set

$$h_n = d_n - R_H(T_H(d_n)).$$

Since $T_H(f_H) = 0$, continuity gives $T_H(d_n) \rightarrow 0$, and boundedness of R_H gives $h_n \rightarrow f_H$. Moreover $T_H(h_n) = 0$, so $h_n \in \mathcal{D}(G) \cap I$. Lemma 2 yields $h_n \in j(C)$, and closedness gives $f_H \in j(C)$.

As H shrinks to the identity, Carter's compact-neighbourhood approximate identity [1, Proposition 2.16(ii)] gives

$$e_H * f * e_H \longrightarrow f \quad \text{in } \mathcal{A}.$$

Thus $f \in j(C)$. We have proved $I \subseteq j(C)$; the reverse inclusion is part of Carter's construction, so $I = j(C)$. $\square$

Corollary 4 (All finite coadjoint-orbit sets). *Let N be a group of unipotent matrices over a p -adic field, and let $\mathcal{A} = L^\Phi(N, \omega)$ satisfy the hypotheses above. Then every finite subset of*

$$\mathfrak{n}^* / \text{Ad}^*(N) \cong \text{Prim}_*(\mathcal{A})$$

is a set of synthesis.

Proof. Carter's Theorem 8.10 states that N is CCR and identifies its unitary dual with the coadjoint-orbit space. If $H \leq N$ is compact-open and π is irreducible, then $\pi(e_H)$ belongs to $\pi(C^*(N)) = \mathcal{K}(\mathcal{H}_\pi)$. It is also an orthogonal projection, hence has finite rank. Thus every irreducible representation is admissible. The CCR property also makes primitive ideals maximal, so the primitive spectrum is T_1 and every finite subset is closed. Apply Theorem 3 and Carter's Corollary 8.11. $\square$

6 The retained abelian compact-open criterion

Theorem 5. *If G is abelian, every compact-open subset of the Pontryagin dual $\widehat{G} \cong \text{Prim}_*(\mathcal{A})$ is a set of synthesis.*

Proof. Fourier transformation maps $\mathcal{D}(G)$ onto $\mathcal{D}(\widehat{G})$. If a compact-open $U \subseteq \widehat{G}$ is disjoint from a closed C , let $e_U \in \mathcal{D}(G)$ satisfy $\widehat{e_U} = 1_U$. Scale e_U to $h = ae_U$ with $0 < a < 2\pi$ and $\|h\|_1 \leq 1$. Every representation in C annihilates h . A Carter cutoff which is zero near 0 and one at a gives $\varphi\{h\} = e_U \in m(C)$. Hence every test function whose Fourier support is contained in U belongs to $j(C)$.

Now let C itself be compact-open and choose $q_C \in \mathcal{D}(G)$ with $\widehat{q_C} = 1_C$. For $f \in \ker(C)$, Fourier injectivity gives $f * q_C = 0$. If $f_n \in \mathcal{D}(G)$ tends to f , put

$$g_n = f_n - f_n * q_C.$$

Then

$$\|g_n - f\|_{\mathcal{A}} \leq (1 + \|q_C\|_{\mathcal{A}}) \|f_n - f\|_{\mathcal{A}} \longrightarrow 0,$$

and the compact-open support of $\widehat{g_n}$ is disjoint from C . The cutoff argument gives $g_n \in j(C)$, hence $f \in j(C)$. Therefore $\ker(C) = j(C)$. $\square$

7 Verification, limitations, and novelty

The proof uses no numerical computation. Its critical points were checked independently: compact-open indicators are the source’s approximate identity; CCR turns their represented projections into finite-rank projections; the right inverse is used only after fixing H , so no uniform bound as $H \downarrow \{e\}$ is required; and the central-support projection belongs to Carter’s exact generating set $m(C)$.

The result does not classify infinite subsets of a nonabelian orbit space. Local indexes and bounded arXiv searches on 21 July 2026 used the exact source id and combinations of “spectral synthesis”, “ p -adic unipotent”, “finite sets”, “singleton”, “primitive spectrum”, and “weighted Orlicz”. No exact prior theorem was found. Carter explicitly notes that even the unweighted $L^1(N)$ case would be novel [1, Section 8.3]; novelty confidence is nevertheless only moderate pending expert review.

Human-review recommendation. Check Lemma 2, especially the central support inside the finite Hecke algebra, and check the equality $T_H(\mathcal{D}_H) = T_H(\mathcal{A}_H)$ before the finite-dimensional lift is chosen. These are the only new structural steps.

References

- [1] M. Carter, *Weighted Orlicz $\ast$ -algebras on locally elliptic groups*, *Studia Math.* **287** (2026), no. 2, 145–180; arXiv:2503.20735v2.